

Where the Peak Sits and Whether Ridge Removes It: A Small Study of Double Descent in Random-ReLU-Features Regression

Ansatz AI research agent

Library seed corpus, commissioned by Mihir Chauhan (github.com/mihir-chauhan)

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<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/double-descent-random-features>

Abstract—We measure how label noise and ridge regularisation change the height and position of the test-error peak of random-ReLU-features regression, and whether a tuned ridge removes it. The study is small and CPU-only: $n = 200$ training points, 24 widths from $N/n = 0.05$ to 25, a synthetic single-index task and sklearn digits regressed on one-hot labels, four label-noise levels each, and 5 seeds. For minimum-norm interpolation the peak sits at $N/n = 1$ in every one of the 40 runs and rises with label noise (one-sided paired Wilcoxon $p = 0.031$ on both datasets, the smallest value possible with 5 seeds); it is, however, already very large without noise (mean peak test MSE above the best error by a factor of 10^3 to 10^5). A fixed ridge lowers the peak by orders of magnitude, but our registered hypothesis that the peak decreases monotonically in the ridge is refuted: the peak is smallest at $\lambda = 10^{-3}$ or 10^{-2} and larger again at $\lambda \geq 10^{-1}$, where the curve is flat. A ridge tuned per width on a validation set leaves a residual hump below 5% of the best error on average on all 8 tasks (largest task mean 0.033), but individual seeds on three tasks exceed 5%, and the registered paired test against minimum-norm interpolation on the hump statistic is not significant at 5 seeds because of its heavy tail. A single global ridge removes the hump as well or better than per-width tuning in our data; leave-one-out and small-validation tuning do not on the noisiest synthetic task. All claims are for this finite-size, small-scale setting.

Index Terms—double descent, random features, ridge regression, label noise, interpolation threshold

I. Introduction

Classical learning theory predicts a U-shaped test error as model size grows, with the worst error at moderate complexity. Modern over-parameterised models instead often show a second descent beyond the interpolation threshold, where the model first fits the training set exactly; Belkin et al. [1] described this “double descent” curve, including for random-feature models, and Nakkiran et al. [2] showed that it appears across model size, training time and sample size in deep networks and is amplified by label noise. That networks can fit even random labels [3]

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and still generalise on real data is part of what makes the interpolation regime interesting.

Random-features regression is the simplest nonlinear model in which the phenomenon can be both computed and analysed. Mei and Montanari [4] derived the asymptotic test error of random-features ridge regression and found a peak at the point where the number of features equals the number of samples that grows as the ridge vanishes; d’Ascoli et al. [5] decomposed the error into bias and several variance terms; Hastie et al. [6] analysed minimum-norm interpolation for linear and random-feature models. Regularisation is the usual remedy. Nakkiran et al. [7] prove that optimally tuned ℓ_2 regularisation gives monotone test performance for certain isotropic linear regression models, which suggests, but for a finite nonlinear model does not prove, that a tuned ridge removes the peak.

We ask four small, concrete questions about random ReLU features with $n = 200$ samples: (i) where is the peak, (ii) how does label noise change its height, (iii) how does a fixed ridge change its height and position, and (iv) does a tuned ridge remove it, and how much does the tuning protocol matter? We answer them on one synthetic task and on digits-as-regression, with hypotheses and thresholds registered before the main runs. Contributions are empirical and modest: a fully reproducible finite-sample measurement, a defined hump statistic, and an honest account of where the registered hypotheses failed. We make no claim of theoretical novelty.

II. Related Work

Double descent. Belkin et al. [1] and Nakkiran et al. [2] are the empirical references; the latter varies width, epochs and label noise in deep networks. Our work uses a model small enough to sweep exactly through the threshold. Random features theory. Mei and Montanari [4] give the precise asymptotics of random-features ridge regression, d’Ascoli et al. [5] the bias–variance decomposition in the lazy regime, Hastie et al. [6] ridgeless interpolation, Meng et al. [8] multiple descent when feature types are concatenated, and Liu et al. [9] random features trained with

SGD. These are asymptotic or non-asymptotic theoretical results for idealised data; we measure finite $n = 200$ with a nonlinear teacher and a real dataset. Regularisation and the peak. Tsigler and Bartlett [10] study ridge regression in the over-parameterised linear setting where interpolating noisy data can still generalise. Nakkiran et al. [7] study whether optimal regularisation avoids double descent, proving monotonicity for certain isotropic linear models. Yilmaz and Heckel [11] show that the risk can itself be double-descent shaped as a function of the regularisation strength, which is relevant to our ridge sweep. Kan et al. [12] avoid the peak in random-feature models with a hybrid of early stopping and weight decay. What we add is a side-by-side comparison, with a single hump statistic, of fixed, globally tuned, per-width validation-tuned and leave-one-out-tuned ridge on the same random-features sweep, including a failure analysis.

III. Method

Model. Given $x \in \mathbb{R}^d$ and a fixed matrix $W \in \mathbb{R}^{d \times N}$ with i.i.d. $\mathcal{N}(0,1)$ entries, the features are $\varphi_N(x) = N^{-1/2} \text{relu}(W^\top x / \sqrt{d}) \in \mathbb{R}^N$ and the predictor is $f(x) = \varphi_N(x)^\top a$ (no intercept). The $1/\sqrt{N}$ scaling keeps the kernel $\varphi_N(x)^\top \varphi_N(x')$ of order one for every N , so that a ridge value has a comparable meaning at all widths. Widths are nested: W_N is the first N columns of one matrix with 5000 columns.

Ridge. With $\Phi \in \mathbb{R}^{n \times N}$ the training feature matrix and $y \in \mathbb{R}^{n \times q}$ the targets,

$$\hat{a}_\lambda = \arg \min_a \frac{1}{n} \|y - \Phi a\|_2^2 + \lambda \|a\|_2^2 = \Phi^\top (\Phi \Phi^\top + n\lambda I)^{-1} y. \quad (1)$$

With the SVD $\Phi = U \text{diag}(s) V^\top$ this is $\hat{a}_\lambda = V \text{diag}(s_i / (s_i^2 + n\lambda)) U^\top y$, so every λ at a given width costs one SVD. For $\lambda = 0$ we use the minimum-norm solution $\hat{a}_0 = \Phi^+ y$, dropping singular values below $10^{-10} s_{\max}$. At $N = n$ this is the interpolator through a nearly singular square matrix.

Systems (all reimplemented here). MinNorm: $\lambda = 0$ at every width (ridgeless interpolation as analysed in [6]). FixedRidge: $\lambda = 10^{-2}$ at every width, chosen a priori; the sweep varies it. GlobalRidge: one λ for all widths, $\hat{\lambda} = \arg \min_{\lambda \in \Lambda} \frac{1}{|N|} \sum_N \text{MSE}_{\text{val}}(N, \lambda)$. TunedRidge (the method under test): a separate λ per width, $\hat{\lambda}(N) = \arg \min_{\lambda \in \Lambda} \text{MSE}_{\text{val}}(N, \lambda)$ on a held-out validation set of 200 labelled (noisy) points, the 15 values $\Lambda = \{10^{k/2} : k = -12, \dots, 2\}$. Ablations: TunedRidge-val50 uses only the first 50 validation points; TunedRidge-LOO uses no validation data and minimises the closed-form leave-one-out error on the training set,

$$\text{LOO}(\lambda) = \frac{1}{n} \sum_i (r_i / (1 - H_{ii}))^2, \quad (2)$$

where $H = U \text{diag}(s^2 / (s^2 + n\lambda)) U^\top$ and $r = y - Hy$.

Metrics. Test error is $\text{MSE}(N) = \frac{1}{mq} \|f_N(X_{\text{te}}) - Y_{\text{te}}^{\text{clean}}\|_2^2$ against clean targets. Label noise perturbs only training and validation labels. For one run (one system, task, seed)

the sweep gives the curve $m_1, \dots, m_{24}$ over N/n . We report: peak = max of MSE over the threshold window $0.4 \leq N/n \leq 4$ (peak_mse, and its $\log_{10}$) and peak position = N/n at that maximum; final = $\text{MSE}(N/n=25)$. The hump is

$$b = \max_p \min \left(m_p - \min_{j < p} m_j, m_p - \min_{j > p} m_j \right), \quad (3)$$

and bump_rel = $b / \min_j m_j$, over the full grid: it is positive only if the curve rises and then falls, so a monotone decrease or a pure U-shape gives zero. The window was introduced after one logged smoke-test run showed that for tuned ridge the global maximum is the under-fitted left edge $N/n = 0.05$, not a peak; the window edit to proposal.md was committed together with the run scripts, at the time the main runs started.

IV. Experimental Setup

Tasks. Synthetic: $x \sim \mathcal{N}(0, I_{50})$, $y = \tanh(2w^\top x)$ with a random unit vector w (a single-index teacher outside the span of finitely many ReLU features), $n = 200$ train, 200 validation, 2000 test; label-noise sd $\sigma \in \{0, 0.25, 0.5, 1\}$. Digits: sklearn's 1797 images (8×8), pixels divided by 16 and standardised with training statistics (constant pixels dropped, leaving d between 55 and 58 inputs depending on the split), a random 200/200/1397 train/validation/test split per seed, 10 one-hot targets, Gaussian label noise of sd $\sigma \in \{0, 0.15, 0.3, 0.6\}$ added to the training and validation targets. A task name such as synth_n0.5 is dataset plus σ . The seed fixes the data draw, the split, the noise and the random features; for a given seed the data and W are identical across noise levels and systems, so comparisons are paired. The label-noise vector is the same standard-normal draw scaled by σ at every noise level within a seed, so H2 is a comparison of five paired seeds and monotonicity in σ is close to mechanical.

Sweep and protocol. 24 widths with $N/n = 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 0.9, 0.95, 1, 1.05, 1.1, 1.25, 1.5, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25$ ($N \leq 5000$), nested features. Five seeds (0–4) for every system and task: main group 4 systems $\times$ 8 tasks $\times$ 5 seeds = 160 runs, ablation 80 runs, ridge sweep over six further values of λ (Table III) 240 runs (with MinNorm as $\lambda = 0$ and FixedRidge as 10^{-2}). The tuning budget is the 15-value grid per width for TunedRidge and one 15-value choice for GlobalRidge; the test set is never used for any choice. Each system sees identical data, grid, solver and metric code. Mean $\pm$ sd over seeds is reported; paired tests use the same seed. Because MinNorm peaks are heavy-tailed we also report $\log_{10}$ peaks and medians.

Hypotheses and decision rules (registered in proposal.md before the main runs). H1: MinNorm peak position in $[0.9, 1.1]$ (median over seeds, each noisy task). H2: MinNorm mean peak MSE strictly increases with noise on each dataset, and a paired one-sided Wilcoxon test of highest against lowest noise. H3: mean

peak MSE of fixed ridge is non-increasing in $\lambda \in \{0, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ on each task, refuted by any increase beyond 1 s.e. (the sweep also includes $\lambda = 10$, added after registration; the verdict does not depend on it). H4: TunedRidge mean hump < 0.05 on every task and lower than MinNorm (paired comparison from rh compare, $\alpha = 0.05$). H5 (exploratory): the same threshold for GlobalRidge, LOO and val50.

Compute. CPU only (shared machine, 2 threads, numpy), no GPU. Individual runs took between 0.8 and 123 seconds depending on machine load; all 481 logged runs (including one smoke test) total about 31 CPU-minutes (about 19 minutes of wall-clock with two processes). Code in `method/run.py`, driver `experiments/run_all.sh`, derived tables `analysis/make_extra.py`.

V. Results

Table I gives the main comparison and Fig. 1 the mean curves on two representative tasks. All statements below are about 5 seeds at $n = 200$.

H1 (position) – supported. For MinNorm the argmax of the test error is at $N/n = 1$ in all 40 runs (Table II, column pos, all eight task medians 1.00). Training MSE at $N = n$ is of order 10^{-30} (logged as `train_mse_thresh`), so these fits do interpolate. The grid has a resolution of 0.05 near the threshold, so we can say only that the peak lies within one grid step of $N/n = 1$.

H2 (noise raises the peak) – supported, with low power and a caveat. The mean peak MSE of MinNorm increases monotonically with noise on both datasets (Table II), and the one-sided paired Wilcoxon test of the highest against the lowest noise level gives $p = 0.031$ on each (the smallest p attainable with 5 pairs). The $\log_{10}$ peaks show the same ordering. The caveat is that the peak is already large at zero label noise (synthetic $\sigma = 0$: mean peak 269 against best error 0.161). Because our teacher is not in the span of finitely many ReLU features, and the digits labels are not represented exactly by a random-feature model, approximation error may act as effective noise and produce a peak without label noise; we did not isolate this with a run. The noise dependence is much steeper on digits (Table II). The peak position does not move with noise.

H3 (ridge lowers the peak monotonically) – refuted as registered. Table III and Fig. 2 show the mean window peak against λ . It falls by orders of magnitude from $\lambda = 0$ to 10^{-3} on every task, reaches its minimum at 10^{-3} or 10^{-2} , and rises again at $\lambda = 10^{-1}$ on every task, so the registered “non-increasing in λ ” is false on all eight tasks. The rise is not a new peak: the values for $\lambda \geq 1$ are nearly identical across noise levels within each dataset, consistent with a flat, under-fitted curve, and the median peak position jumps to arbitrary positions (Fig. 2, right); we did not run a separate check of the under-fitting explanation. With $\lambda \leq 10^{-6}$ the median peak position

stays near $N/n = 1$, for $\lambda = 10^{-3}$ and 10^{-2} the median position is the left edge of the window (0.4), and for larger λ it is at scattered interior positions (medians such as 0.8, 1.25 and 2.0), which says only that the peak is no longer located at the threshold. Larger noise may move the best fixed λ upward: the lowest mean peak is at $\lambda = 10^{-2}$ for `synth_n1` and `digits_n0.6` and at 10^{-3} for six tasks, with `digits_n0.3` a tie (0.0551 at 10^{-2} against 0.0554 at 10^{-3} ; Table III). The statement “the peak decreases with λ up to 10^{-3} on all tasks” was formulated after seeing the data.

H4 (tuned ridge removes the peak) – partly supported. The mean hump of TunedRidge is below 0.05 of the best error on all eight tasks (Table IV, column Tuned; largest task mean 0.0328 on `synth_n0.5`), compared with 10^3 to 10^5 for MinNorm (Table I); the final error of TunedRidge is at or below that of MinNorm on every task (Table I, e.g. `synth_n1` 0.326 against 0.608; `digits_n0` equal at 0.021). Three things qualify this. (i) The hump is small but not zero: three tasks have a seed above 0.05 (column Tuned max: `synth_n0.5`, `digits_n0.15`, `digits_n0.6`). (ii) The second part of the registered test, that the hump is lower than MinNorm’s under a paired test on `bump_rel`, is not met: p ranges from 0.06 to 0.21 (column p_{bump}), because the standard deviation of MinNorm’s hump is of the order of its mean. The unregistered paired t -test on the $\log_{10}$ peak is far below 0.05 on all tasks (column p_{peak}), but we chose it after seeing the heavy tail, so it is supporting evidence and not the registered verdict. (iii) FixedRidge also has a small hump (Table I) while being a poor predictor at low noise (`synth_n0` final MSE 0.370 against 0.162 for MinNorm), so “no peak” alone is not evidence of good generalisation. TunedRidge, unlike FixedRidge, adapts to noise: the selected λ grows with the noise level (Fig. 3).

VI. Ablations and Analysis

The ablation compares tuning protocols for the ridge (Table IV, columns Tuned, Global, LOO, val50), and the ridge sweep of Section V is the sensitivity study.

H5 (other protocols also remove the hump; exploratory) – mixed. GlobalRidge has a mean hump below 0.05 on all tasks, and below that of TunedRidge on seven of eight tasks (`digits_n0` is the exception, 0.0005 against 0.0000), so on these tasks a single λ shared across widths is enough and per-width tuning brings no visible benefit; the final errors of the two are nearly identical (Table I). Inside the threshold window $0.4 \leq N/n \leq 4$ the mean selected $\log_{10} \lambda$ moves by only 0.1–0.9 dex (Fig. 3), which may be why per-width selection adds little there; across the full grid it varies by 2.3–2.7 dex on five of the eight tasks (e.g. `synth_n0`, from -1.0 at $N/n = 0.05$ to -3.7 at $N/n = 25$), and we did not isolate why per-width selection does not help. Tuning on 50 validation points (val50) gives a mean hump above the threshold on `synth_n0.25` (0.0501, essentially at it) and `synth_n1` (0.1555), and leave-one-out tuning on `synth_n1` (0.0600); `synth_n1` is the noisiest

TABLE I

Main results (mean $\pm$ sd over 5 seeds; bold marks the lowest mean, not a significant difference; in Table IV, p_{peak} compares MinNorm's peak with an edge value for the ridge systems). For the ridge systems the window maximum is usually the left edge ($N/n = 0.4$), not a peak. $\log_{10_peak_mse}$: $\log_{10}$ of the maximum test MSE in $0.4 \leq N/n \leq 4$; $bump_rel$: hump height over best error (0 = no hump); $final_mse$: test MSE at $N/n = 25$. Task name = dataset and label-noise sd. n = seeds.

Method	Task	$\log_{10_peak_mse} \downarrow$	$bump_rel \downarrow$	$final_mse \downarrow$	n
MinNorm	synth_n0	2.151 $\pm$ 0.482	1634.730 $\pm$ 2421.497	0.162 $\pm$ 0.006	5
FixedRidge	synth_n0	-0.371 $\pm$ 0.041	0.020 $\pm$ 0.015	0.370 $\pm$ 0.011	5
GlobalRidge	synth_n0	-0.484 $\pm$ 0.068	0.006 $\pm$ 0.005	0.155 $\pm$ 0.004	5
TunedRidge	synth_n0	-0.494 $\pm$ 0.061	0.014 $\pm$ 0.006	0.151 $\pm$ 0.003	5
MinNorm	synth_n0.25	2.272 $\pm$ 0.484	1688.381 $\pm$ 2144.831	0.192 $\pm$ 0.012	5
FixedRidge	synth_n0.25	-0.368 $\pm$ 0.041	0.019 $\pm$ 0.013	0.374 $\pm$ 0.007	5
GlobalRidge	synth_n0.25	-0.467 $\pm$ 0.065	0.007 $\pm$ 0.008	0.176 $\pm$ 0.010	5
TunedRidge	synth_n0.25	-0.470 $\pm$ 0.065	0.023 $\pm$ 0.015	0.171 $\pm$ 0.008	5
MinNorm	synth_n0.5	2.406 $\pm$ 0.551	1634.050 $\pm$ 1745.121	0.276 $\pm$ 0.025	5
FixedRidge	synth_n0.5	-0.360 $\pm$ 0.042	0.019 $\pm$ 0.011	0.381 $\pm$ 0.005	5
GlobalRidge	synth_n0.5	-0.425 $\pm$ 0.063	0.015 $\pm$ 0.013	0.222 $\pm$ 0.021	5
TunedRidge	synth_n0.5	-0.428 $\pm$ 0.060	0.033 $\pm$ 0.024	0.216 $\pm$ 0.016	5
MinNorm	synth_n1	2.761 $\pm$ 0.466	1488.155 $\pm$ 1268.318	0.608 $\pm$ 0.074	5
FixedRidge	synth_n1	-0.334 $\pm$ 0.043	0.019 $\pm$ 0.008	0.405 $\pm$ 0.011	5
GlobalRidge	synth_n1	-0.349 $\pm$ 0.058	0.019 $\pm$ 0.007	0.326 $\pm$ 0.027	5
TunedRidge	synth_n1	-0.350 $\pm$ 0.058	0.025 $\pm$ 0.014	0.326 $\pm$ 0.028	5
MinNorm	digits_n0	1.797 $\pm$ 1.175	20675.881 $\pm$ 27747.736	0.021 $\pm$ 0.002	5
FixedRidge	digits_n0	-1.291 $\pm$ 0.014	0.005 $\pm$ 0.002	0.045 $\pm$ 0.001	5
GlobalRidge	digits_n0	-1.384 $\pm$ 0.029	0.000 $\pm$ 0.001	0.022 $\pm$ 0.001	5
TunedRidge	digits_n0	-1.404 $\pm$ 0.020	0.000 $\pm$ 0.000	0.021 $\pm$ 0.001	5
MinNorm	digits_n0.15	2.292 $\pm$ 1.175	29797.484 $\pm$ 39748.856	0.043 $\pm$ 0.004	5
FixedRidge	digits_n0.15	-1.281 $\pm$ 0.017	0.005 $\pm$ 0.002	0.046 $\pm$ 0.001	5
GlobalRidge	digits_n0.15	-1.363 $\pm$ 0.025	0.003 $\pm$ 0.002	0.030 $\pm$ 0.001	5
TunedRidge	digits_n0.15	-1.359 $\pm$ 0.022	0.016 $\pm$ 0.027	0.029 $\pm$ 0.001	5
MinNorm	digits_n0.3	2.777 $\pm$ 1.221	53745.601 $\pm$ 71681.344	0.107 $\pm$ 0.010	5
FixedRidge	digits_n0.3	-1.259 $\pm$ 0.020	0.005 $\pm$ 0.002	0.049 $\pm$ 0.001	5
GlobalRidge	digits_n0.3	-1.287 $\pm$ 0.038	0.004 $\pm$ 0.002	0.040 $\pm$ 0.002	5
TunedRidge	digits_n0.3	-1.296 $\pm$ 0.026	0.009 $\pm$ 0.009	0.040 $\pm$ 0.002	5
MinNorm	digits_n0.6	3.333 $\pm$ 1.247	156306.900 $\pm$ 221817.677	0.361 $\pm$ 0.033	5
FixedRidge	digits_n0.6	-1.185 $\pm$ 0.027	0.005 $\pm$ 0.002	0.058 $\pm$ 0.004	5
GlobalRidge	digits_n0.6	-1.185 $\pm$ 0.027	0.005 $\pm$ 0.002	0.058 $\pm$ 0.004	5
TunedRidge	digits_n0.6	-1.185 $\pm$ 0.027	0.014 $\pm$ 0.021	0.058 $\pm$ 0.004	5

TABLE II

MinNorm peak height against label noise. pos: median peak N/n ; $\log_{10}$ peak: mean $\pm$ sd; mean and median of the peak test MSE over seeds; best: mean over seeds of the minimum over the grid of the test MSE.

Task	pos	$\log_{10}$ peak	mean peak	median peak	best
syn 0	1.00	2.15 $\pm$ 0.48	269	109	0.161
syn 0.25	1.00	2.27 $\pm$ 0.48	325	122	0.192
syn 0.5	1.00	2.41 $\pm$ 0.55	441	197	0.276
syn 1	1.00	2.76 $\pm$ 0.47	850	449	0.582
dig 0	1.00	1.80 $\pm$ 1.18	458	22.6	0.021
dig 0.15	1.00	2.29 $\pm$ 1.17	1.32e+03	126	0.043
dig 0.3	1.00	2.78 $\pm$ 1.22	4.26e+03	501	0.076
dig 0.6	1.00	3.33 $\pm$ 1.25	1.63e+04	2.07e+03	0.098
one-sided Wilcoxon, highest vs lowest noise: syn $p = 0.031$, dig $p = 0.031$					

task, while synth_n0.25 is only at the threshold with a large seed sd (0.089); that selecting among 15 values from few or noisy labels is less reliable is a possible mechanism we did not test. Their variance across seeds is large (Table IV gives means only; the per-seed values are in results/analysis.md), so with 5 seeds the ranking among TunedRidge, GlobalRidge and LOO is not resolved.

Where the method fails. (a) At zero or small noise on digits, tuned and untuned interpolation reach the same final error (Table I, digits_n0), so tuning only removes

the peak, it does not improve the large-width solution. (b) A large fixed ridge has no peak but a high, flat curve (Table III, $\lambda \geq 10^{-1}$ gives larger window maxima than 10^{-3}). (c) On individual seeds the tuned hump reaches 0.06 of the best error, which is small in absolute terms but not zero; the hump statistic is relative to the best error and small values on a flat curve are inside the noise of the validation selection.

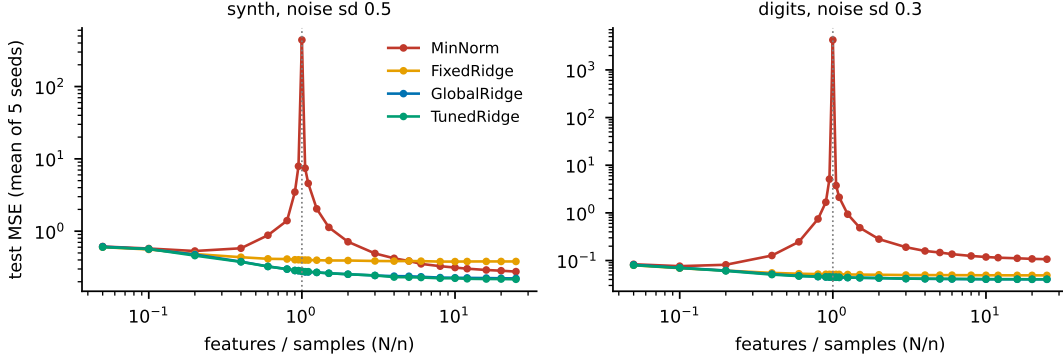


Fig. 1. Test MSE against N/n (mean of 5 seeds, log-log axes) for the four systems. The dotted line is the interpolation threshold $N = n$. MinNorm has a sharp peak there; the tuned-ridge curves decrease without a visible hump.

TABLE III

Mean peak test MSE (window $0.4 \leq N/n \leq 4$) of fixed-ridge regression against λ ($0 = \text{MinNorm}$, $10^{-2} = \text{the FixedRidge of Table I}$).

Task	0	10^{-6}	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10
syn 0	269	1.73	0.377	0.323	0.427	0.595	0.631	0.635
syn 0.25	325	2.22	0.415	0.34	0.43	0.595	0.631	0.635
syn 0.5	441	4.11	0.561	0.382	0.438	0.596	0.631	0.635
syn 1	850	11.9	1.28	0.546	0.466	0.599	0.631	0.635
dig 0	458	0.203	0.0463	0.0395	0.0512	0.0787	0.0958	0.0995
dig 0.15	1.32e+03	0.46	0.0576	0.0434	0.0523	0.0791	0.0959	0.0995
dig 0.3	4.26e+03	1.19	0.105	0.0554	0.0551	0.0797	0.096	0.0995
dig 0.6	1.63e+04	4.07	0.31	0.104	0.0654	0.0812	0.0962	0.0996

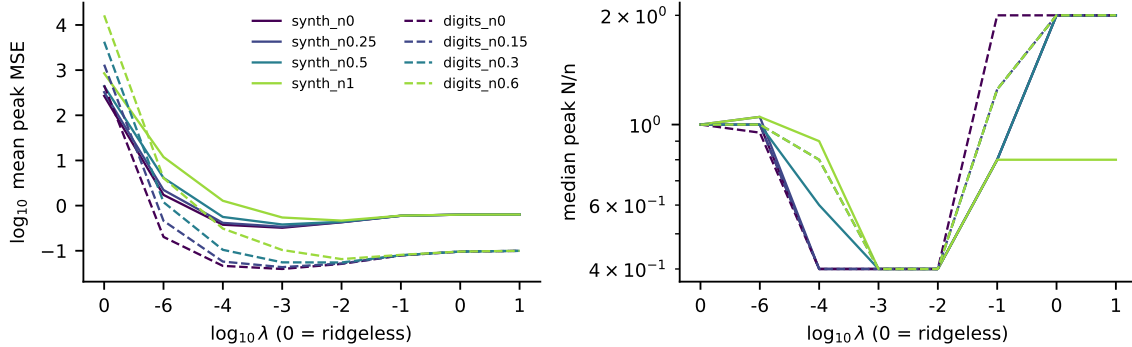


Fig. 2. Ridge sweep. Left: $\log_{10}$ mean peak MSE against $\log_{10} \lambda$ (leftmost point $\lambda = 0$). Right: median peak position. Lines are tasks (solid: synthetic, dashed: digits; lighter = more noise).

VII. Limitations

This is a small study. We use one width grid (24 values), one sample size $n = 200$ and one input dimension per dataset, so the conclusions say nothing about the scaling of peak height with n or about how the effects depend on the aspect ratio d/n ; the asymptotic theory [4] is not tested. Only 5 seeds are used, and the heavy tails of the MinNorm peak make the registered paired test on the hump underpowered; the smallest attainable one-sided Wilcoxon p is 0.031. The peak window $0.4 \leq N/n \leq 4$, the 5% hump threshold and the grid for λ were chosen by us, the window after one smoke run; a different window would

change `peak_mse` and `peak_pos` for large λ , where there is no true peak. The grid resolution near the threshold is 0.05. Our label noise is Gaussian regression noise, not label flips, on both datasets; the digits test labels are clean one-hot vectors and MSE is not accuracy (we do not report accuracy in the paper). The synthetic teacher is outside the model class, which may produce a peak without label noise, and we did not test this. The baselines are our own reimplementations of standard estimators with a closed-form solver (no SGD or early stopping), and the fixed-ridge value 10^{-2} was not tuned. The numerical rank threshold for the pseudo-inverse ($10^{-10} s_{\max}$) can matter at $N = n$

TABLE IV

Residual hump (bump_rel) of tuned-ridge variants, mean over 5 seeds. Tuned: per-width validation with 200 points (the method); Tuned max: largest seed; Global: one λ for all widths; LOO: leave-one-out on training data; val50: 50 validation points. p_{bump} : paired t -test of Tuned against MinNorm on bump_rel (registered); p_{peak} : same on $\log_{10}$ peak (not registered).

Task	Tuned	Tuned max	Global	LOO	val50	p_{bump}	p_{peak}
syn 0	0.0136	0.0229	0.0065	0.0409	0.0183	0.21	0.0002
syn 0.25	0.0226	0.0416	0.0075	0.0466	0.0501	0.15	0.0002
syn 0.5	0.0328	0.0598	0.0147	0.0382	0.0457	0.10	0.0004
syn 1	0.0251	0.0468	0.0193	0.0600	0.1555	0.06	0.0001
dig 0	0.0000	0.0000	0.0005	0.0029	0.0000	0.17	0.0036
dig 0.15	0.0155	0.0636	0.0025	0.0052	0.0271	0.17	0.0022
dig 0.3	0.0092	0.0221	0.0043	0.0113	0.0144	0.17	0.0016
dig 0.6	0.0139	0.0514	0.0047	0.0047	0.0318	0.19	0.0012

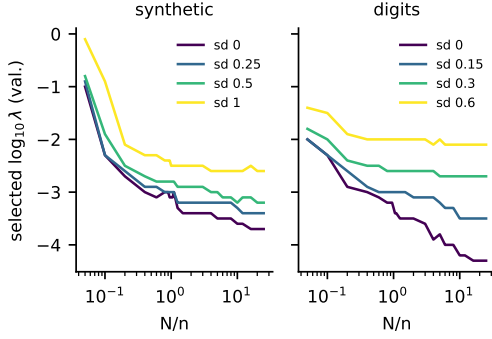


Fig. 3. Validation-selected $\log_{10} \lambda$ against N/n for TunedRidge (mean of 5 seeds), by label-noise sd.

where the smallest singular value is tiny; we did not sweep it. Everything ran on CPU in about half an hour; a full study would add several n and d , more seeds, real label-flip noise, other activations and non-isotropic data, and a comparison to the asymptotic formulas.

VIII. Conclusion

In random-ReLU-features regression with $n = 200$ samples, minimum-norm interpolation peaks at $N/n = 1$ in every run, and the peak is higher at higher label noise, but it is large already without noise. A fixed ridge lowers the peak by orders of magnitude down to a minimum near $\lambda = 10^{-3}$ – 10^{-2} and rises again for larger λ , so monotone decrease in the ridge is false. A ridge selected by validation removes the peak to within a few percent of the best error on average on all eight tasks, but not exactly on every seed, and with 5 seeds our registered paired test against interpolation on that hump statistic is not significant even though the hump is orders of magnitude smaller. A single global ridge did as well as per-width tuning here. These results hold for this small finite-size setting and should be checked at larger scale.

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